

ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

UNIT: COMPUTER NETWORKING AND DESIGN

CONTINUOUS ASSESSMENT TEST (CAT) 3

TIME: 2 HOURS
TOTAL MARKS: 50

NAME: _____

ADMISSION NO: _____

INSTRUCTIONS

1. Answer ALL questions.
2. Show all workings where applicable.
3. Write your answers clearly and neatly.

QUESTION ONE (20 MARKS)

1. Differentiate between the **OSI Model** and the **TCP/IP Model**. (2 Marks)
2. State any two functions of the **Network Layer** in the OSI model. (2 Marks)
3. Define the following terms: (2 Marks)
 - a) IPv4 Address
 - b) IPv6 Address
4. Identify the class of each of the following IP addresses: (2 Marks)
 - a) 10.0.0.1
 - b) 172.16.5.10
5. Differentiate between: (2 Marks)

- a) Static IP Address
- b) Dynamic IP Address

6. State any three advantages of IPv6 over IPv4. (3 Marks)

7. Explain the term **CIDR (Classless Inter-Domain Routing)**. (2 Marks)

8. State any two advantages of subnetting. (2 Marks)

9. Differentiate between a **Switch** and a **Router**. (3 Marks)

QUESTION TWO (15 MARKS)

10. Outline three steps involved in the initial setup and configuration of a Cisco router. (3 Marks)

11. Write the Cisco IOS commands used to: (3 Marks)

- a) Enter global configuration mode.
- b) Assign an IP address to an interface.
- c) Save the running configuration.

12. Explain the purpose of the following routing protocols: (3 Marks)

- a) RIP
- b) OSPF
- c) EIGRP

13. State any two benefits of implementing VLANs in a network. (2 Marks)

14. What is an Access Control List (ACL) and why is it used in network security? (2 Marks)

15. Differentiate between NAT and PAT. (2 Marks)

QUESTION THREE (15 MARKS)

16. State any two benefits of implementing Port Security on a switch. (2 Marks)

17. Explain the purpose of Spanning Tree Protocol (STP) in a switched network. (2 Marks)

18. What is Link Aggregation Control Protocol (LACP) and why is it used? (2 Marks)

19. Explain any three configurations performed when setting up a Wireless Access Point (AP). (3 Marks)

20. State the functions of the following network services: (2 Marks)

- a) DHCP
- b) DNS

21. What is an SSID and why is it important in wireless networking? (2 Marks)

22. Describe two measures that can be used to secure a wireless network. (2 Marks)

23. Define the term **Network Privileges** and state one example. (2 Marks)

QUESTION FOUR: WIRESHARK AND NETWORK SECURITY (10 MARKS)

24. Using Wireshark packet captures, explain the **TCP Three-Way Handshake** by describing the following packets: (4 Marks)

- a) SYN
- b) SYN-ACK
- c) ACK
- d) State the purpose of the TCP Three-Way Handshake.

25. What is Wireshark and state two uses of Wireshark in network administration? (3 Marks)

26. A network administrator wants to display only TCP packets in Wireshark: (3 Marks)

- a) Write the display filter used. (1 Mark)
- b) State one advantage of packet analysis during network troubleshooting. (2 Marks)

TOTAL: 50 MARKS

END OF PAPER